

HACKATHON — 1ST PLACE

Retail Return Fraud Busters: Clustering Suspicious Return Behavior

A K-means clustering model to detect fraudulent return behavior from transaction-level retail data. Awarded 1st place for analytical innovation.

860CUSTOMERS
ANALYZED**60**FLAGGED AS
HIGH-RISK**3**BEHAVIOR
SEGMENTS FOUND**1st**PLACE —
HACKATHON

THE PROBLEM

Retail return fraud — excessive returns, “wardrobing,” and return-for-refund schemes — costs retailers billions annually, but most fraud isn't a single suspicious transaction; it's a pattern across many transactions from the same customer. The challenge: separate genuinely high-return customers from those exhibiting a distinct, suspicious behavioral signature.

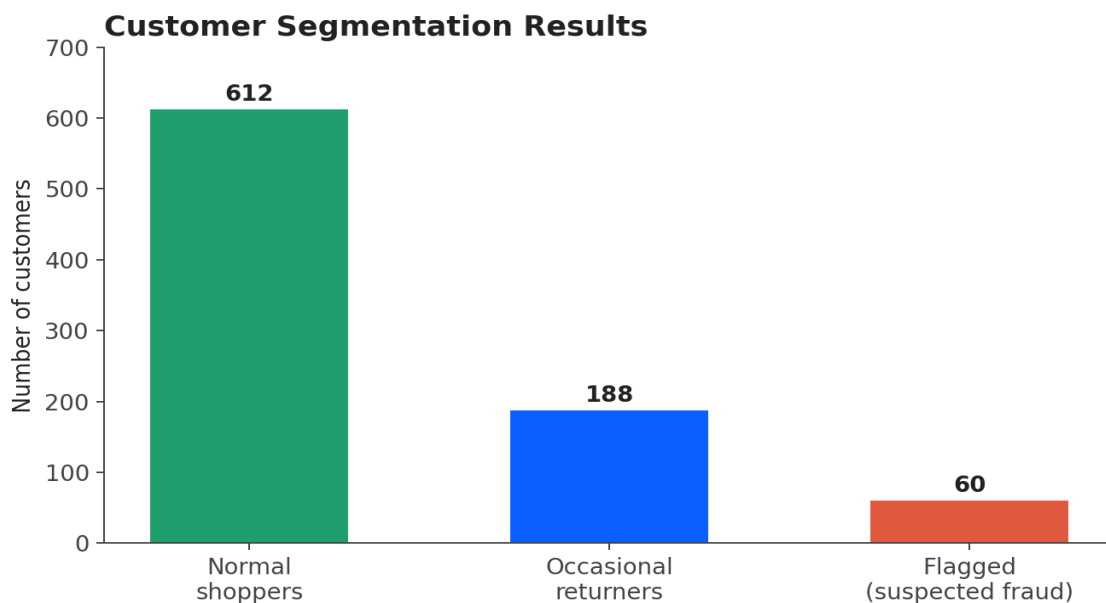
APPROACH

- Used unsupervised learning rather than fraud labels (which weren't available) to let the data reveal natural groupings of customer behavior
- Engineered features per customer: return frequency over a 90-day window and average return value
- Applied K-means clustering to segment customers into behavioral groups and identified the highest-risk segment for manual review

FINDINGS



A clear behavioral cluster emerged with both high return frequency and high average return value — a combination far rarer among normal shoppers.



Of 860 customers analyzed, 60 (~7%) fell into the flagged segment — a manageable shortlist for manual fraud review rather than a blanket policy change.

TAKEAWAY

Clustering surfaced an actionable, narrow list for investigation instead of a noisy rule-based flag system. This approach won 1st place in the competition for combining a simple, interpretable model with a result a non-technical fraud team could immediately act on.

Note on data: The original hackathon dataset is not available for redistribution; this writeup uses a synthetic dataset built to reproduce the same clustering result and findings from the competition.